Functions and Average Rate of Change

Algebra II Practice — 45 Minutes

Name: _____ Date: _____

Instructions

Complete all 12 problems in 45 minutes without a calculator. You may use this cover sheet. Show key algebraic steps and give exact answers. Use interval notation for domains and ranges. For conceptual questions, give a mathematical reason or a counterexample; a yes/no answer alone is not sufficient.

Average Rate of Change (ARC)

For $a < b$ and defined endpoint values $f(a)$ and $f(b)$,

$$\boxed{\mathrm{ARC}_{[a,b]}(f) = \frac{f(b) - f(a)}{b - a}}.$$

This is the slope of the secant line through $(a, f(a))$ and $(b, f(b))$; its units are output units per input unit. The difference quotient $\dfrac{f(x+h) - f(x)}{h}$ requires $h \neq 0$ and both function values to exist.

ARC for a Quadratic on $[a, b]$

Let $f(x) = Ax^2 + Bx + C$, where $A \neq 0$. Since $b^2 - a^2 = (b - a)(a + b)$,

$$\mathrm{ARC}_{[a,b]}(f) = \frac{A(b^2 - a^2) + B(b - a)}{b - a} = \frac{(b - a)\big(A(a + b) + B\big)}{b - a}.$$

$$\boxed{\mathrm{ARC}_{[a,b]}(f) = A(a + b) + B} \qquad (a < b).$$

In vertex form, $f(x) = A(x - h)^2 + k$, the equivalent formula is $\mathrm{ARC}_{[a,b]}(f) = A(a + b - 2h)$.

Function Foundations

Functions and notation

A function assigns exactly one output to each input in its domain. The domain is the set of allowed inputs; the range is the set of actual outputs. The graph of a function passes the vertical line test. The notation $f(x)$ is a function value, not multiplication by f.

Domain restrictions

Denominators cannot be zero. An even-root radicand must be nonnegative. A real logarithm argument must be positive. Preserve restrictions from the original expression even after algebraic simplification. For piecewise functions, use the branch whose condition contains the input.

Composition

$$(f \circ g)(x) = f(g(x)).$$

Its domain requires $x \in \mathrm{Dom}(g)$ and $g(x) \in \mathrm{Dom}(f)$.

One-to-one functions and inverses

One-to-one means distinct inputs have distinct outputs; the graph passes the horizontal line test. An inverse function requires a one-to-one original function, possibly after restricting its domain.

$$f^{-1}(f(x)) = x \quad \text{on } \mathrm{Dom}(f).$$

Inverse functions exchange domain and range. The notation f^{-1} means inverse, not reciprocal.

Functions and Average Rate of Change — Practice

1. Consider the relation

$$R = \{(-3, 4),\ (-1, 2),\ (0, 4),\ (2, -1)\}.$$

State its domain and range. Determine whether R is a function and whether its inverse relation is a function. Justify both decisions.

2. Let $f(x) = \dfrac{\sqrt{5 - x}}{x - 2}$. Find its natural real domain. Find $f(-4)$ and $f(5)$, and explain why $f(2)$ and $f(6)$ are not defined as real function values.

3. For $q(x) = -2x^2 + 7x - 3$, find

$$\mathrm{ARC}_{[-1,4]}(q), \qquad \mathrm{ARC}_{[1,3]}(q), \qquad \mathrm{ARC}_{[t,t+h]}(q) \quad (h > 0).$$

Give the last answer in terms of t and h.

Functions and Average Rate of Change — Continued

4. Let $f(x) = x^2 - 6x + 8$. Find every interval $[a, b]$ of length 4 for which $\mathrm{ARC}_{[a,b]}(f) = 0$. Is f constant on the interval you found? Justify your answer using function values.

5. A quadratic $p(x) = Ax^2 + Bx + C$ satisfies

$$\mathrm{ARC}_{[0,2]}(p) = 5, \qquad \mathrm{ARC}_{[1,4]}(p) = 11, \qquad p(0) = -3.$$

Determine A, B, and C.

6. The table gives the water remaining in a tank.

t (minutes)	0	1	3	6
$V(t)$ (liters)	24	20	9	3

Find the average rates of change on $[0, 3]$ and $[1, 6]$, including units. A student says, "The tank lost exactly 5 liters during each minute from $t = 0$ to $t = 3$." Does the table justify that claim? Explain.

Functions and Average Rate of Change — Continued

7. Let

$$p(x) = \begin{cases} x^2 + 1, & x < 1, \\ 3x - 1, & 1 \le x < 4, \\ 10 - x, & x \ge 4. \end{cases}$$

Find $p(1)$, $p(4)$, $\mathrm{ARC}_{[-1,4]}(p)$, and $\mathrm{ARC}_{[1,4]}(p)$.

8. Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 - 3$. Find simplified formulas and domains for $f \circ g$ and $g \circ f$. Then find $\mathrm{ARC}_{[1,5]}(g \circ f)$.

9. Let $q(x) = (x-3)^2 - 4$. Explain why q has no inverse function on its full real domain, using two specific inputs. Restrict its domain to $[3, \infty)$ and find $q^{-1}(x)$, together with the domain and range of that inverse.

Functions and Average Rate of Change — Continued

10. Decide whether each statement is true or false. Give a reason for each true statement and a counterexample for each false statement.

    (a) If $\mathrm{ARC}_{[a,b]}(f) > 0$, then f must be strictly increasing throughout $[a, b]$.

    (b) If two functions with domain $\mathbb{R}$ have equal ARC on every interval $[a, b]$, then they must be the same function.

    (c) A function can have two distinct inputs with the same output.

    (d) If both compositions exist, then $f \circ g = g \circ f$.

11. Consider

$$f(x) = \frac{x^2 - 9}{x - 3}, \qquad g(x) = x + 3,$$

    each with its natural real domain. Are these the same function? Explain. Compute their average rates of change on $[-1, 2]$. Does agreement of those rates establish that the functions are equal?

12. For $f(x) = 2x^2 - 3x + 1$ and $h > 0$, a student claims

$$\frac{f(a + h) - f(a)}{h} = \frac{f(h)}{h}.$$

    Determine whether the claim holds for all real a and positive h. Derive the correct simplified expression for the left side and give a specific counterexample if the claim is false. Also explain the difference between $f(a + h)$ and $f(a) + h$.